

# Jacob Wong

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## Education

Queen's University, B.Eng in Computer Engineering (Co-op) September 2024 - April 2028

Dean's List (Fall 2024, Winter 2025, Fall 2025)

**Relevant courses:** Data Science, Data Structures and Algorithms, Object-Oriented Programming, Economics

## Work Experience

**Data Engineering Intern** May 2026 - August 2026

Royal Bank of Canada Toronto, ON

- Built the **Python framework** for an **agent-driven workflow** that joins mutual fund data from S3 files (CSV, JSON, Parquet) with **Trino** tables across **4** scenarios, **saving 4 hours** weekly.
- Orchestrated **Spark** jobs and **SQL** queries with **Airflow DAGs**, completing **25** consecutive runs without failure.
- Verified agent-driven workflow outputs by writing a **PyTest** suite with **SQL** checks that confirmed data existed after each run, reaching **95%** coverage and validating results in **Jupyter Notebook**.

**Autonomy Software Lead** July 2026 - Present

Queen's Aerospace Design Team Kingston, ON

- Led a team of **9** students building **PX4** and **ROS2** autonomy for SAE Aero2027, earning an **11x** point multiplier.
- Automated drone imagery uploads to **GCP** through an **ETL** pipeline, sustaining **1 MB/s** (3.6 GB per flight hour).
- Containerized **ROS2** nodes with **Docker**, cutting developer setup from **4 hours to 1** across a **10-person** team.
- Set GitHub review standards for the team, catching **15+** bugs across **5** pull requests before merge.

**Engineering Instructor** June 2025 - August 2025

STEM Camp Toronto, ON

- Taught **Python** with **Tkinter** and **PyGame** to **150+** youth, raising average test scores by **20%**.
- Built a Micro:bit pedometer streaming live sensor data, matching manual step counts with **90%** accuracy.

## Projects

NHL Game Stats ELT pipeline | *Kafka, Databricks, AWS S3, Google Cloud Platform, BigQuery* [GitHub](#)

- Built an **ELT pipeline**, replaying NHL game event data via **Kafka** into **S3**, simulating a live game of **300+** events.
- Developed a **Databricks** medallion process that ingests raw play-by-play JSON from **S3** and refines it through bronze, silver, and gold **Delta tables**, producing goal, assist, and shot totals, supporting **35+** skaters.
- Wrote a **Python** script that loads season stats for any 2 teams (up to **40 skaters**) into **BigQuery** on **GCP**, creating a table per request, finishing each load in under **10** seconds.
- Assembled a **Streamlit** GUI that checks game and live skater totals against the expected stats.

MoveModel | *Python, pandas, matplotlib, NumPy* [GitHub](#)

- Engineered a **logistic regression** model to classify walking versus jumping, reaching **95%** test accuracy.
- Cleaned **3** accelerometer axes in **pandas**, removing **20%** noisy readings before training.
- Plotted predictions in **matplotlib**, showing **85%** of errors were walking misclassified as jumping.

Distracted Driving Monitor | *Python, OpenCV, Raspberry Pi, Git* [GitHub](#)

- Built an **OpenCV** distracted-driving detector, recognizing **90%** of distraction signs across **50+** test clips.
- Logged distraction scores to **Excel** with **Python** across **10** driver sessions, enabling time-based review.

## Technical Skills

**Programming Languages:** Python, SQL, C, C++, Java, Bash

**Data and Pipelines:** Spark, Airflow, Kafka, Confluent, Databricks, Delta Lake, Trino, BigQuery, ETL/ELT

**Cloud and DevOps:** AWS S3, Google Cloud Platform, Docker, Git, GitHub Actions, Linux, Jira

**Libraries and Tools:** pandas, NumPy, matplotlib, PyTest, Jupyter, OpenCV